

1. The technique of integration used to find the indefinite integral $\int x^2 e^{-x} dx$ is

- A. partial fractions B. by parts C. trigonometric substitution D. power rule

2. If $x > 1$, and $y = \int_1^{x^2} \sqrt{1 + \sqrt{t}} dt$, then $\frac{dy}{dx}$ is equal to

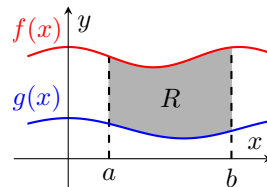
- A. $1 + x$ B. $2x\sqrt{1 + x}$ C. $\sqrt{1 + \sqrt{x}}$ D. $\sqrt{1 + x^2}$

3. The length of the arc on the graph of $y = f(x)$ from the point $(a, f(a))$ to the point $(b, f(b))$ is computed using

- A. $\int_a^b \sqrt{1 + [f'(x)]^2} dx$ B. $\int_a^b \sqrt{1 + [f(x)]^2} dx$ C. $\pi \int_a^b \sqrt{1 + \frac{d^2y}{dx^2}} dx$ D. $\pi \int_a^b (f(x))^2 dx$

4. The integral expression which gives the area of the shaded region R in the figure is

- A. $\int_a^b (g(x) - f(x)) dx$ B. $\int_a^b (f(x) - g(x))^2 dx$
C. $\pi \int_a^b (f(x) - g(x)) dx$ D. $\int_a^b (f(x) - g(x)) dx$



5. Let $f(x)$ be continuous on $[-5, 5]$ and $\int_0^5 f(x) dx = 13$. If for each x in $[-5, 5]$, $f(-x) = f(x)$, i.e., $f(x)$ is an even function, then $\int_{-5}^5 f(x) dx =$

- A. 26 B. 13 C. 0 D. -13

6. Let $f(x)$ be continuous on $[-5, 5]$ and $\int_{-5}^0 f(x) dx = 13$. If for each x in $[-5, 5]$, $f(-x) = -f(x)$, i.e., $f(x)$ is an odd function, then $\int_{-5}^5 f(x) dx =$

- A. 26 B. 13 C. 0 D. -13

7. The average value of a function f on the interval $[a, b]$ is given by

- A. $\frac{1}{b-a} \int_a^b f(x) dx$ B. $(b-a) \int_a^b f(x) dx$ C. $\frac{b}{a} \int_a^b f(x) dx$ D. $\frac{f(b) - f(a)}{b - a}$

8. If $\int_1^2 f(x) dx = 5$, then $\int_1^2 (f(x) + 2x) dx$ is equal to

A. 5

B. 8

C. 10

D. $2x \cdot f(x)$

9. If $\int_b^a f(x) dx = 13$, then $\int_a^b f(x) dx =$

A. 13

B. 6.5

C. -13

D. 0

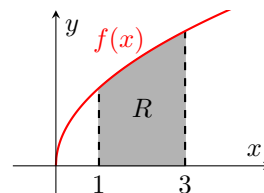
10. The volume of the solid generated by revolving the shaded region R about the x -axis is computed using the formula

A. $\pi \int_1^3 (f(x))^2 dx$

C. $\int f(x) dx$

B. $\pi \int_1^3 f(x) dx$

D. $\pi \int_1^3 \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$



11. $\int_3^{\sqrt{3}} \frac{x^4 + x^2 + 1}{\sqrt{x^8 + x^4 + 2}} dx$

A. 9

B. 0

C. 6

D. $\frac{1}{\sqrt{3}}$

Find each indefinite integral for problems 12 through 16. *Show your work.*

12. $\int \frac{dx}{(x-1)(x+3)}$

13. $\int xe^{-x} dx$

14. $\int \frac{dx}{9+x^2}$

15. $\int \frac{2x}{9+x^2} dx$

16. $\int \tan^5 x \sec^2 x dx$

17. Find the area of the region bounded by the graphs of $y = x^2 + 2$ and $y = x - 2$ from $x = -1$ to $x = 1$.

18. Find the volume of the solid generated by revolving the region bounded by the the x -axis and the curve $y = \sqrt{x+1}$ from $x = 0$ to $x = 3$ around the x -axis.

19. Find the length of the arc on the graph of $y = \frac{2}{3}x^{3/2} + 9$ from the point $(0, 9)$ to the point $(3, 2\sqrt{3} + 9)$.

20. If $F(x)$ is an antiderivative of $f(x)$ on the interval $[a, b]$ and $F(a) = 2$, $F(b) = 5$ then $\int_a^b f(x) dx =$

A. 7

B. 5

C. 2

D. 3

21. Solve the initial value problem, $\frac{dy}{dx} = \sqrt{x+1}$ and $y(0) = 1$.

22. Evaluate $\int_{-1}^3 |x-2| dx$.

23. Use the figure to evaluate $\int_{-4}^4 f(x) dx$.

